

CleanTrack: End-to-End Workflow Specification & Comprehensive Module Actions Catalog

A Complete Operational and Technical Reference Detailing the Closed-Loop Civic Cleanliness Workflow, AI-Powered Waste Triage Pipeline, Dynamic SLA Escalations, and Exhaustive User Action Inventories Across Citizen, Municipal Officer, Field Sanitation Squad, and City Administrator Modules.

System Application: CleanTrack Urban Waste Operating System • **Version:** 2.4.0 (Production Build)
Target Domain: Municipal Solid Waste (MSW), Biomedical Hazard Segregation & E-Waste Tracking
Architecture Layer: Multi-Tenant Role-Based Access Control (RBAC), Asynchronous Event-Driven Microservices
Underlying AI Engine: YOLOv8 Deep Vision Ensemble + ViT Feature Extractor + Perceptual Duplicate Hash
Statutory Framework: Swachh Bharat Mission (Urban 2.0), CPCB Solid & Biomedical Waste Guidelines

Executive Summary & Operational Paradigm: CleanTrack solves the critical breakdown in traditional municipal grievance portals where complaints sit in unmonitored backlogs without actionable intelligence. CleanTrack implements an unbroken, automated digital chain-of-custody: *Citizen Photo Ingestion* → *Instant Multi-Modal Computer Vision Triage* → *50-Meter Spatial Duplicate Merging* → *Dynamic Risk-Weighted SLA Generation* → *Municipal Resource & Machine Allocation* → *Worker Route-Optimized Dispatch* → *Live Geo-Tagged After-Photo Proof* → *Supervisory Verification* → *Citizen Loop Closure with Gamified Green Points*. Every module features explicit, role-gated action flows designed for zero-friction civic engagement and strict municipal accountability.

Document Architecture & Report Structure

Section	Coverage & Technical Scope	Key Artifacts & Diagrams
Section 1: Master Workflow	End-to-end 9-phase operational lifecycle from initial photo snap to citizen rating and long-term recurrence mitigation.	Closed-loop process pipeline & phase specifications.
Section 2: Citizen Actions	Complete user catalog: camera capture, GPS reverse geocoding, AI preview, tracking, resolution audit, and Green Points.	9 granular citizen actions with step-by-step UI flows.
Section 3: Municipal Actions	Triage officer actions: priority queues, spatial duplicate merging, recurring dumps, GIS mapping, squad dispatch, and QA.	11 municipal staff actions with decision matrices.
Section 4: Field Worker Actions	Sanitation squad actions: shift check-in, hazard briefing, navigation, clearance execution, and geo-tagged proof photo upload.	8 field execution actions with safety checklists.
Section 5: Admin & Public	Commissioner oversight, ward benchmarking, user RBAC, AI monitoring, public transparency, and automated background daemons.	City command oversight & background cron engine.
Section 6: State Lifecycle	Cross-module state transition matrix, event triggers, failure handling, and full database entity models.	State machine table & schema reference.

1. Master End-to-End Civic Cleanliness Workflow

The CleanTrack operational lifecycle establishes an automated, closed-loop civic governance pipeline. Unlike legacy portals that treat citizen submissions as unstructured static text tickets, CleanTrack passes every report through computer vision triage, spatial clustering, dynamic SLA computation, workforce dispatch, and supervisory quality verification before issuing civic reward tokens.

Phase & Name	Primary Actor	Technical Process & System Operations	SLA / Output State
Phase 1: Ingestion & Pre-Flight	Citizen / App	Citizen captures live photo or selects file. Browser extracts EXIF metadata and GPS coordinates. HTML5 Geolocation API initiates reverse geocoding to resolve street address, zone, and ward ID.	Status: <i>Pre-submission</i> EXIF + GPS Pin resolved
Phase 2: AI Multi-Modal Inference	AI Vision Engine	Image streamed to YOLOv8 inference service. Performs bounding box localization, multi-class waste taxonomy classification (Plastics, Biomedical Sharps, E-Waste, Organic), volume estimation, and hazard index scoring.	Confidence score (e.g. 94%) Hazard index calculated
Phase 3: Spatial Duplicate Clustering	Geospatial Engine	MongoDB \$geoNear spatial query performs 50-meter radius search for unresolved complaints. Perceptual image hashing calculates visual similarity. If duplicate detected, report is linked into primary cluster.	Duplicate linked or Single incident confirmed
Phase 4: SLA Binding & Prioritization	Priority Service	Multi-factor formula calculates Composite Priority Score: $\text{Score} = 0.35(\text{WasteType}) + 0.25(\text{Volume}) + 0.20(\text{Sensitivity}) + 0.15(\text{Recurrence}) + 0.05(\text{Time})$. Generates hard SLA due date (CRITICAL: 12h, MEDIUM: 24h, NORMAL: 48h).	Status: AI Analyzed dueAt timestamp locked
Phase 5: Municipal Triage & Dispatch	Triage Officer	Complaint appears in Municipal Priority Queue sorted by SLA urgency. Officer evaluates AI recommendations for workforce crew size, machinery (JCB / mini-dumper), and PPE kits, then assigns Sanitation Squad.	Status: Assigned Worker alerted via push
Phase 6: Field Worker Clearance	Field Squad	Squad receives work order on mobile device. Inspects hazard warning briefing, triggers turn-by-turn GPS navigation, marks status 'In Progress', performs physical clearance and segregation, and transports waste.	Status: In Progress Physical clearance executed
Phase 7: Geo-Tagged Proof Upload	Squad Lead	Squad captures live 'After' photo at the site. Camera module enforces live GPS validation and timestamping to prevent fraudulent gallery re-uploads. Logs disposal transfer station and volume cleared.	Status: Awaiting Verification Proof uploaded
Phase 8: Dual-Photo Verification & QA	Municipal Supervisor	Supervisor inspects side-by-side Before/After comparison slider and AI cleanliness confidence score. One-click approval resolves the ticket; rejection returns ticket to field squad with required remediation notes.	Status: Resolved or Rejected Formal QA sign-off
Phase 9: Citizen Closure & Gamification	Citizen / Platform	Citizen receives real-time notification with resolution evidence. Citizen verifies clean site and submits 1-5 star rating. System credits Green Points to citizen account, updating eco-tier rank.	Status: Citizen Verified +50 Green Points issued

Automated SLA Escalation Engine (Background Cron Service): Running continuously via node-cron every 15 minutes, the escalation daemon queries all active complaints where $\text{NOW} > \text{sla.dueAt}$ and status is not yet 'Resolved'. Overdue tickets are automatically promoted to **CRITICAL**, an immutable entry is logged in priorityHistory, and emergency WebSocket alert banners are pushed directly to Zonal Municipal Officers and the City Commissioner.

2. Citizen Module: Detailed User Actions & System Flows

Module Persona & Scope: Designed for everyday citizens, resident welfare associations, and community volunteers. Focuses on zero-friction waste reporting, real-time AI transparency, active complaint tracking, and civic gamification incentives. Primary Route Prefix: /citizen/*.

Action Code & Name	User Interaction / Steps	System Processing & AI Logic	Resulting State & Output
ACT-C01: Live Camera / Photo Upload	1. Taps 'Report Waste' button. 2. Clicks camera icon to trigger device camera or selects existing photo from storage.	Validates file type (JPEG/PNG/WebP, max 10MB). Strips malicious payloads, extracts EXIF orientation, generates local base64 thumbnail preview.	Image preview rendered on screen; ready for AI scanning.
ACT-C02: Geolocation & Landmark Pin	1. Device prompts for GPS permission. 2. Citizen confirms detected pin or drags marker on interactive Leaflet map to adjust exact garbage dump spot. 3. Enters nearby landmark.	Browser HTML5 Geolocation queries hardware coordinates. Reverse geocoder translates lat/long into ward name (e.g. 'Ward 12 - Shivaji Nagar') and street address. MongoDB verifies ward boundary polygon.	GeoJSON [longitude, latitude] stored; Ward ID auto-assigned.
ACT-C03: Real-Time AI Pre-Analysis	1. Observes automatic scanning animation over uploaded photo. 2. Views detected waste categories, bounding boxes, and confidence score.	Dispatches asynchronous request to YOLOv8 inference endpoint. Analyzes pixel features; classifies waste stream (Plastics, Hazardous Medical, E-Waste); estimates severity and volume.	AI Confidence Badge (e.g. 94%), Segregation Stream, and Severity tag displayed.
ACT-C04: Register & Submit Complaint	1. Verifies AI detected category (or selects override dropdown if needed). 2. Adds optional textual notes. 3. Toggles 'Report Anonymously' if desired. 4. Clicks 'Submit Official Grievance'.	Generates unique tracking code (CT-2026-XXXX). Executes duplicate check query. Inserts document into MongoDB complaints collection with initial SLA timer.	Status set to AI Analyzed. Success modal shown with Complaint ID and SLA countdown.
ACT-C05: Track Active Grievances	1. Navigates to 'My Complaints' tab. 2. Filters complaints by status (All, In Progress, Resolved). 3. Observes real-time progress bars.	Retrieves user grievances sorted by submission timestamp. Computes remaining time until SLA breach: remainingTime = dueAt - NOW. Updates live via WebSocket listeners.	Color-coded SLA countdown badge (Green = Safe, Red = Imminent Breach).
ACT-C06: Inspect Resolution Evidence	1. Clicks on a resolved grievance ticket. 2. Drags interactive Before/After comparison slider. 3. Views clearance timestamp and squad notes.	Fetches paired images: original imageUrl vs worker proof workerEvidence.imageUrl. Displays cleanup audit log and disposal transfer station details.	Visual confirmation of restored site; enables citizen feedback.
ACT-C07: Citizen Verification & Rating	1. Taps 'Confirm Cleanliness' button. 2. Selects 1 to 5 star rating for field squad. 3. Submits satisfaction remarks.	Patches complaint status to Citizen Verified. Logs citizen satisfaction score. Dispatches trigger to Reward Engine to credit civic incentive tokens.	+50 Green Points credited instantly; Thank You badge awarded.
ACT-C08: Explore Ward Hotspots Map	1. Opens 'Citizen Hotspots' view. 2. Inspects ward-level cleanliness index heatmap. 3. Clicks nearby community reports to upvote.	Aggregates open complaint clusters within 2km radius. Renders density circles (Red = Heavy dumping, Green = Clean zone). Upvote increments priority weight.	Community visibility of chronic dumping zones; prevents duplicate filings.
ACT-C09: Redeem Green Points & Perks	1. Visits 'Green Points' ledger. 2. Views current balance, eco-rank tier, and history. 3. Selects reward voucher (e.g. 10% Municipal Tax rebate or Metro transit pass).	Verifies available point balance. Deducts redeemed points from users ledger. Generates cryptographic QR discount coupon code.	Redeemable QR code voucher generated; Tier status updated.

3. Municipal Staff Module: Triage, Dispatch & Quality Control

Module Persona & Scope: Built for Zonal Triage Officers, Municipal Sanitary Inspectors, and Ward Supervisors. Provides decision-support intelligence to triage incoming complaints, merge duplicate reports, deploy heavy machinery, dispatch field squads, and enforce strict quality verification. Primary Route Prefix: /municipal/*.

Action Code & Name	Officer Interaction & Controls	System Processing & Intelligence	Operational Impact
ACT-M01: Priority Queue Triage	1. Opens 'Priority Queue'. 2. Reviews incoming complaints dynamically sorted by SLA urgency and hazard severity. 3. Filters by Ward or Waste Stream.	Queries MongoDB complaints collection with compound index { 'priority.level': 1, 'sla.dueAt': 1 }. Flags complaints under 3 hours to breach with pulsating red visual indicator.	Immediate attention directed to high-risk hazards (biomedical/e-waste).
ACT-M02: Spatial Duplicate Merging	1. Navigates to 'Duplicate Detection' tab. 2. Inspects grouped clusters of complaints. 3. Reviews side-by-side images. 4. Clicks 'Merge into Primary Ticket'.	Algorithm clusters tickets located within 50m with perceptual image similarity > 80%. Merging updates duplicate tickets to merged, aggregates reporter list, and links all updates to parent ticket.	Eliminates redundant truck dispatches to the same garbage spot.
ACT-M03: Recurring Waste Blackspot Audit	1. Opens 'Recurring Waste' page. 2. Reviews chronic blackspot frequency (>3 cleanups in 30 days). 3. Inspects inferred root cause (e.g. missing street bin).	Temporal aggregation queries historical cleanups per coordinate buffer. Calculates recurrence recurrence index. Recommends permanent intervention (CCTV installation or bin deployment).	Shifts municipal strategy from reactive cleaning to root-cause prevention.
ACT-M04: GIS Hotspot & Route Mapping	1. Explores interactive GIS map. 2. Toggles heatmap layers: Waste Volume, Biohazard Density, SLA Overdue zones. 3. Views proximity to sensitive institutions.	Renders Leaflet vector circles with graduated color ramps. Cross-references coordinates with sensitive municipal GIS layers (hospitals, primary schools, water reservoirs).	Strategic spatial visibility of city-wide waste distribution.
ACT-M05: AI Recommendation Review	1. Opens complaint detail. 2. Reviews AI machinery suggestion. 3. Evaluates suggested team size (e.g., 'Requires 4 workers + 1 Mini JCB Loader for construction debris').	Heuristic AI engine cross-references estimated waste volume (m³) and material density. Recommends specialized PPE (heavy-duty puncture-proof gloves for medical sharps).	Prevents worker injury and ensures right-sized equipment dispatch.
ACT-M06: Squad Dispatch & Work Order	1. Selects 'Assign Squad' dropdown. 2. Chooses Sanitation Team (e.g. 'Squad 4 - Central Ward'). 3. Attaches officer instruction note. 4. Clicks 'Confirm Dispatch'.	Creates Assignment record. Updates complaint status to Assigned. Emits WebSocket event TASK_ASSIGNED to field squad tablet/mobile application.	Work order transmitted instantly to field squad mobile device.
ACT-M07: Supervisory Cleanup Verification	1. Opens 'Verification' queue. 2. Reviews worker 'After' photo beside initial 'Before' photo. 3. Checks AI Cleanliness Confidence Score. 4. Clicks 'Approve Resolution'.	System executes dual-image visual difference analysis. If cleanliness score ≥ 85%, green checkmark is displayed. Supervisor approval transitions status to Resolved.	Verified clean; triggers citizen notification and Green Points.
ACT-M08: Rejection & Remediation Re-Dispatch	1. If 'After' photo reveals residual trash or incorrect spot, supervisor clicks 'Reject Proof'. 2. Selects punch list reason (e.g. 'Residual debris left on sidewalk').	Reverts status to In Progress. Appends supervisor feedback to task audit log. Sends high-priority correction push notification to field squad lead.	Enforces 100% thorough cleanup standards before case closure.
ACT-M09: Spam & Invalid Report Triage	1. Reviews 'Invalid Reports' queue. 2. Inspects AI-flagged false submissions (memes, selfies, non-waste images, duplicate spam). 3. Confirms rejection.	YOLOv8 confidence < 25% or zero waste classes automatically flags complaint. Officer rejection marks ticket Rejected and flags user account for potential abuse penalties.	Protects municipal workforce from chasing fake or frivolous reports.
ACT-M10: Municipal Analytics Inspection	1. Visits 'Analytics' dashboard. 2. Reviews ward turnaround times, SLA compliance %, total tonnage cleared, and waste composition breakdowns.	Executes multi-stage MongoDB aggregation pipelines across complaints, assignments, and wards. Computes MTTR (Mean Time to Resolution).	Actionable data for workforce reallocation and performance audits.
ACT-M11: Export AI Intelligence Dossier	1. Opens 'AI Reports' tab. 2. Selects reporting period (Weekly / Monthly) and Ward. 3. Clicks 'Generate Swachh Bharat Compliance PDF'.	Assembles executive analytics, SLA compliance metrics, chronic blackspot audits, and equipment usage logs into publication-ready PDF.	Automated official compliance dossier ready for City Commissioner.

4. Field Worker Module: Task Execution & Evidence Capture

Module Persona & Scope: Engineered for Sanitation Squad Leaders, Waste Collection Drivers, and Field Crew. Designed with high-contrast, touch-first mobile interfaces for rough field conditions, clear hazard warnings, one-tap turn-by-turn navigation, and mandatory geo-tagged post-cleanup verification. Primary Route Prefix: /worker/*.

Action Code & Name	Field Worker Interaction & UI	Background System Actions	Operational Outcome
ACT-W01: Shift Login & Task Roster	- Worker logs in on rugged mobile/tablet. - Reviews assigned tasks for the shift ordered by proximity and SLA urgency.	Filters active assignments where <code>assignedWorkerId == currentWorker</code> and status is not yet 'Resolved'. Highlights critical biohazard tasks.	Clear operational roadmap for the shift; zero ambiguity.
ACT-W02: Task Inspection & Hazard Briefing	- Taps task card. - Inspects initial waste photo, volume estimate, and prominent safety hazard alert (e.g. 'Caution: Medical Sharps Detected').	Displays mandatory safety gear checklist (N95 mask, puncture-resistant gloves, puncture container) before enabling navigation.	Ensures worker occupational safety compliance under CPCB norms.
ACT-W03: Turn-by-Turn GPS Navigation	- Taps 'Navigate to Location' button. - App launches embedded map route or redirects to Google Maps with exact target coordinates.	Calculates real-time driving route from worker's current GPS position to waste coordinates. Displays landmark notes entered by reporting citizen.	Rapid transit directly to garbage spot without searching.
ACT-W04: Check-In & Start Work	- Arrives at dumping location. - Taps 'Start Cleanup Work' button.	Verifies worker GPS coordinates are within 100m geofence of target. Updates complaint status to In Progress. Emits real-time update to citizen.	Citizen sees 'Sanitation Squad has arrived and started cleaning'.
ACT-W05: Physical Cleanup & Segregation	Squad clears waste into designated compartments (Dry recyclable, Wet compostable, Hazardous/Bio-waste).	Operational protocol enforced: hazardous waste placed into yellow/red biohazard containment bins per CPCB standards.	Physical restoration of civic site; source-segregated loading.
ACT-W06: Live 'After' Photo Proof Capture	- Steps back to capture clear photo of restored, clean ground. - Camera module stamps live GPS & time. - Inputs collected waste volume (e.g. '1.5 Tonnes'). - Taps 'Submit Completion'.	Uploads evidence to server storage. Enforces live camera capture (blocks file picker to eliminate fraud). Updates status to Awaiting Verification.	Proof submitted to supervisor verification queue; task completed in field.
ACT-W07: Squad & Equipment Coordination	- Opens 'Squad' tab. - Views team member names, contact numbers, assigned compactor truck ID, and equipment manifest.	Retrieves assigned squad roster and vehicle telematics link. Allows one-tap emergency calling to Zonal Sanitary Inspector.	High cohesion across driver, crew members, and supervisor.
ACT-W08: Performance & Gratitude Ledger	- Opens 'History' tab. - Reviews completed jobs, citizen feedback stars, and monthly productivity score.	Queries historical resolved assignments. Aggregates citizen star ratings. Computes squad completion rate and bonus points.	Worker morale and motivation driven by direct civic appreciation.

5. Administrator & Public Modules: Governance & Transparency

Module Persona & Scope: Tailored for City Municipal Commissioners, Smart City Directors, IT Administrators, and Public Civic Observers. Enforces city-wide governance, cross-ward SLA benchmarking, AI model telemetry, and open data transparency. Primary Route Prefixes: /admin/* and /* (Public).

Action Code & Name	Administrator / Public Flow	System Operation & Telemetry	Governance Outcome
ACT-A01: City Cleanliness Command Center	1. Administrator accesses executive dashboard. 2. Reviews city-wide cleanliness index, total volume processed, active tickets, and breach rate.	Real-time aggregation pipeline computes city-wide MTTR, SLA compliance percentage, and open crisis escalations across all 15 municipal wards.	Bird's eye visibility for Municipal Commissioner.
ACT-A02: Municipal Watch & Benchmarking	1. Opens 'Municipal Watch'. 2. Ranks wards by resolution efficiency and breach rate. 3. Inspects lagging wards.	Computes composite Ward Efficiency Score based on on-time resolution, recurring dump frequency, and citizen satisfaction ratings.	Identifies under-resourced wards needing additional trucks/squads.
ACT-A03: User RBAC & Ward Allocation	1. Opens 'User Management'. 2. Creates/edits user accounts. 3. Assigns roles (Citizen, Staff, Worker, Admin) and ward jurisdictions. 4. Toggles account activation.	Encrypts passwords via bcrypt (10 rounds). Signs JWT tokens with role claims. Enforces strict backend route middleware (authorizeRoles).	Total security and access governance across municipal staff.
ACT-A04: AI Vision Engine Health & Telemetry	1. Visits 'AI Monitoring' console. 2. Inspects model inference latency, confidence distribution histograms, and false positive rates.	Logs latency per inference (average 142ms). Tracks low-confidence edge cases for dataset curation and retraining loop.	Ensures deep learning models operate within production tolerances.
ACT-A05: SLA & System Configuration	1. Opens System Settings. 2. Adjusts SLA hours per priority (Normal: 48h → 36h). 3. Modifies Green Points multiplier.	Updates SystemSetting document. Escalation cron engine and priority calculator dynamically adopt updated parameters without server restart.	Agile policy adjustments during city clean drives and festivals.
ACT-A06: Immutable Security Audit Inspection	1. Opens Audit Log viewer. 2. Filters audit logs by user, date range, or action type (e.g. 'MANUAL_PRIORITY_OVERRIDE').	Queries immutable AuditLog collection recording user ID, IP address, timestamp, previous state, and modified state.	Tamper-proof compliance trail for anti-corruption oversight.
ACT-P01: Public Cleanliness Transparency	1. Any unauthenticated citizen visits public portal. 2. Views live statistics: Tons of waste cleared, active civic campaigns, top wards.	Caches public metric endpoints in Redis/memory to prevent database load. Renders interactive impact counters and environmental metrics.	Builds public trust through radical municipal transparency.
ACT-P02: Interactive AI Intelligence Demo	1. Public visitor opens 'AI Intelligence' page. 2. Toggles sample waste images (PET bottles, syringe sharps, circuit boards) to inspect AI bounding boxes.	Demonstrates real-time browser inference and multi-class segmentation. Educates citizens on proper waste segregation at source.	Promotes community environmental awareness and literacy.

6. Cross-Module State Machine & Technical Specifications

State Transition Protocol: Every complaint entity in CleanTrack transitions through strict, deterministic states. State transitions can only be triggered by authorized actors or verified system daemons, ensuring total integrity.

Status State	Trigger Event & Initiator	Prerequisites & Validation	Automated Notifications & Actions
Reported	Citizen submits form via Web/Mobile.	Valid image uploaded; coordinates within city boundaries.	Complaint ID assigned; image queued for AI inference.
AI Analyzed	AI Vision Service completes inference.	Bounding boxes extracted; confidence $\geq 25\%$.	Priority score computed; dueAt SLA locked; added to queue.
Under Review	Municipal Officer opens ticket details.	Officer possesses valid RBAC role <code>municipal_staff</code> .	Locks ticket for 10 minutes to prevent duplicate assignment.
Assigned	Officer selects Sanitation Squad.	Target squad exists and has active shift status.	WebSocket push to worker mobile; citizen notified of squad name.
In Progress	Worker taps 'Start Work' on site.	Worker GPS within 100m geofence of complaint coordinates.	Citizen receives live notification: 'Squad is cleaning your area'.
Awaiting Verification	Worker submits live 'After' photo.	Photo metadata verified live (camera capture timestamp < 5 min).	Ticket routed to Supervisor Verification Queue; dual-slider loaded.
Resolved	Supervisor approves resolution.	After photo verified clean; AI difference score $\geq 85\%$.	AuditLog stamped; citizen prompted for feedback; SLA timer stopped.
Citizen Verified	Citizen confirms clean & rates squad.	Rating between 1 and 5 stars submitted.	+50 Green Points credited to citizen; squad rating updated.
Rejected	Officer rejects spam / non-waste.	Formal rejection code selected (e.g. 'Invalid Image / Spam').	Citizen notified with formal reason; SLA cancelled; no points awarded.

Mathematical Models & Algorithmic Formulations:

- **Composite Severity Formula:** $Priority_Score = \omega_1 * S_waste + \omega_2 * S_vol + \omega_3 * S_loc + \omega_4 * S_recur + \omega_5 * S_time$, where weights $\omega = [0.35, 0.25, 0.20, 0.15, 0.05]$. Biomedical waste sets $S_waste = 100$.
- **Dynamic SLA Deadline Generation:** $dueAt = createdAt + SLA_Window(Priority_Level)$, where CRITICAL = 12 Hours, MEDIUM = 24 Hours, NORMAL = 48 Hours.
- **Haversine Proximity Clustering:** $d = 2R \cdot \arcsin(\sqrt{(\sin^2(\Delta\phi/2) + \cos(\phi_1)\cos(\phi_2)\sin^2(\Delta\lambda/2))}) \leq 50m$.

Technical Assurance & Deployment Readiness

CleanTrack has been verified under comprehensive end-to-end integration testing across 12 API controllers, 10 MongoDB domain schemas, 4 distinct user persona dashboards, and real-time Socket.io bi-directional event pipelines. The architecture adheres strictly to CPCB waste management compliance, Swachh Bharat Mission 2.0 digital governance protocols, and production-grade microservice security standards.

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SIH Project Domain: Smart Cleanliness & Waste Governance

Build Status: Production Ready (v2.4.0)